Structure of skeletal muscle, muscle tone and excitation-contraction coupling (ECC)

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Learning objectives

- Describe the organization of skeletal muscle from whole muscle to the functional unit(sarcomere).
- Identify the components of sarcomere(Z-line, M-line, A-band, I-band, H-zone)
- Describe the molecular structure of actin and myosin.
- Define motor unit.
- Explain excitation-contraction coupling in skeletal muscles.
- Define muscle tone.

Pre-assessment

Q1. The functional unit of skeletal muscle is:

a) Myofibril
b) Sarcomere
c) Muscle fascicle
d) Myofilament

Q2. Thin filaments are primarily composed of:

a) Myosin
b) Troponin
c) Actin
d) Titin.

Cont..

Q3. Neuromuscular junction uses electrical transmission only?(T/F)

Q4. Acetylcholinesterase breaks down ACh?(T/F)

Muscle introduction

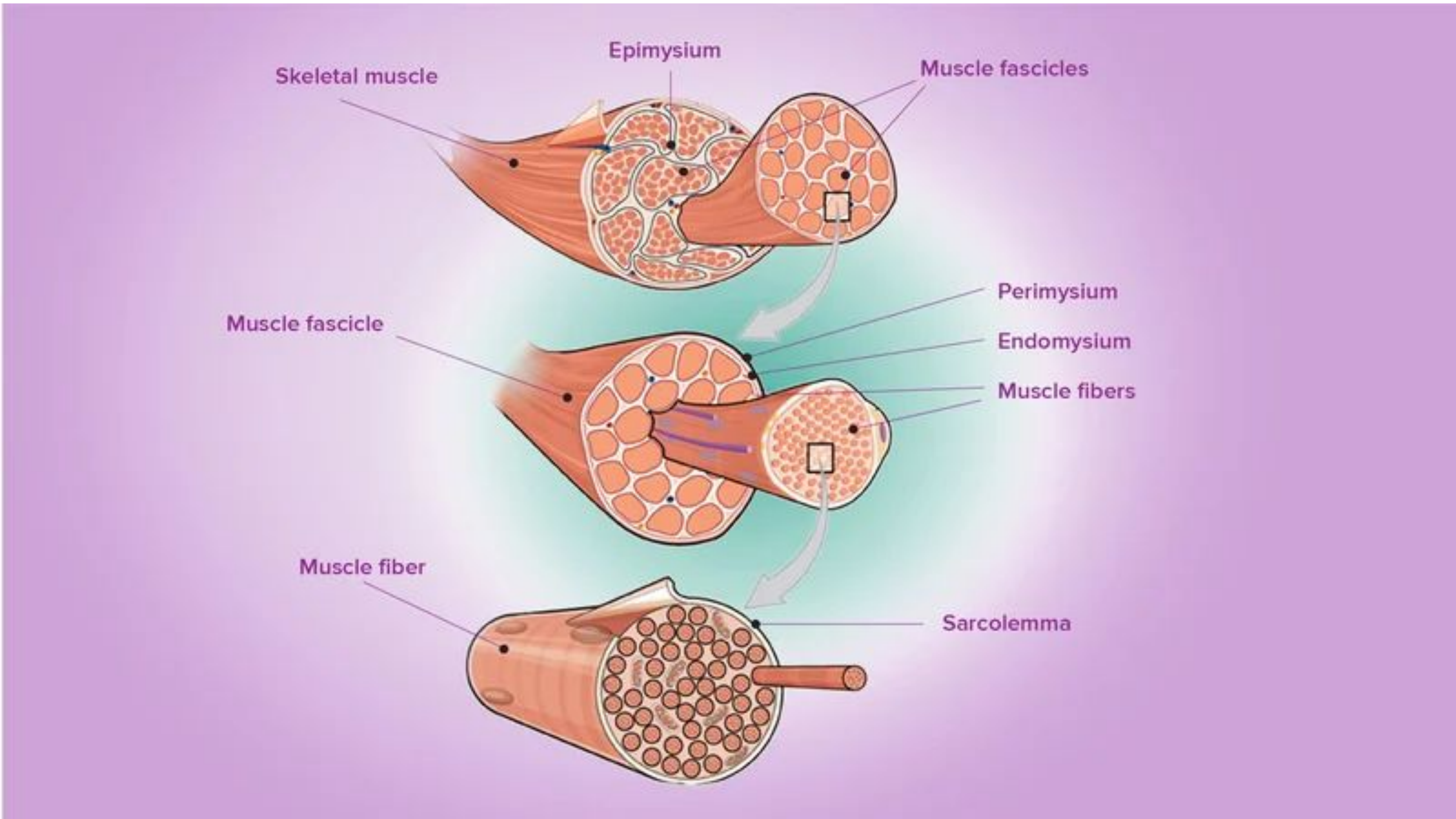
- Muscles are specialized tissues in the body responsible for movement, posture and stability.
- They work by contracting and relaxing.

Human body contains over 600 muscles making up about 40% to 50% of our body weight.

Types of muscles

1. Skeletal muscles (striated, voluntary)
2. Cardiac muscles (striated, involuntary)
3. Smooth muscles (non-striated, involuntary)

MUSCLE-> MUSCLE FASCICLE-> MUSCLE FIBRE-> MYOFIBRIL-> MYOFILAMENTS


Epimysium
Skeletal muscle
Muscle fascicles
Perimysium
Muscle fascicle
Endomysium
Muscle fibers
Muscle fiber
Sarcolemma

Structure of myofibrils

Under microscope the striated appearance is due to alternating appearance of dark and light bands.

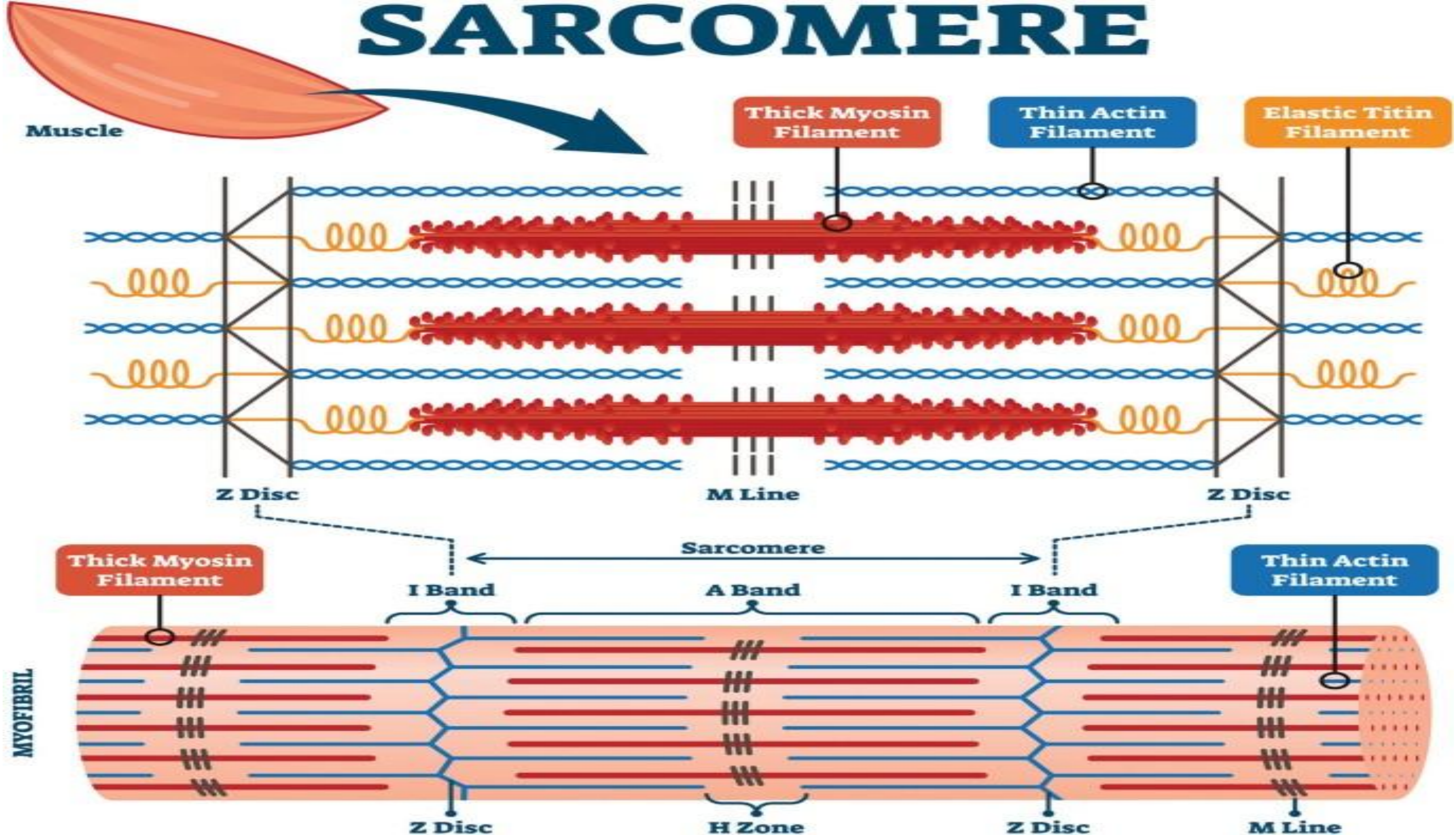
- A- band
- I- band
- Z- Line
- M- line
- H- zone

Sarcomere

Sarcomere is the basic functional unit of muscle fibre.

It is the region b/w two Z-lines.It contains both actin and myosin filaments.

SARCOMERE


Muscle
Thick Myosin Filament
Thin Actin Filament
Elastic Titin Filament
Z Disc
M Line
Z Disc
Sarcomere
Thick Myosin Filament
Thin Actin Filament
I Band
A Band
I Band
MYOFIBRIL
Z Disc
H Zone
Z Disc
M Line

Actin (Thin filament)

It is composed of:

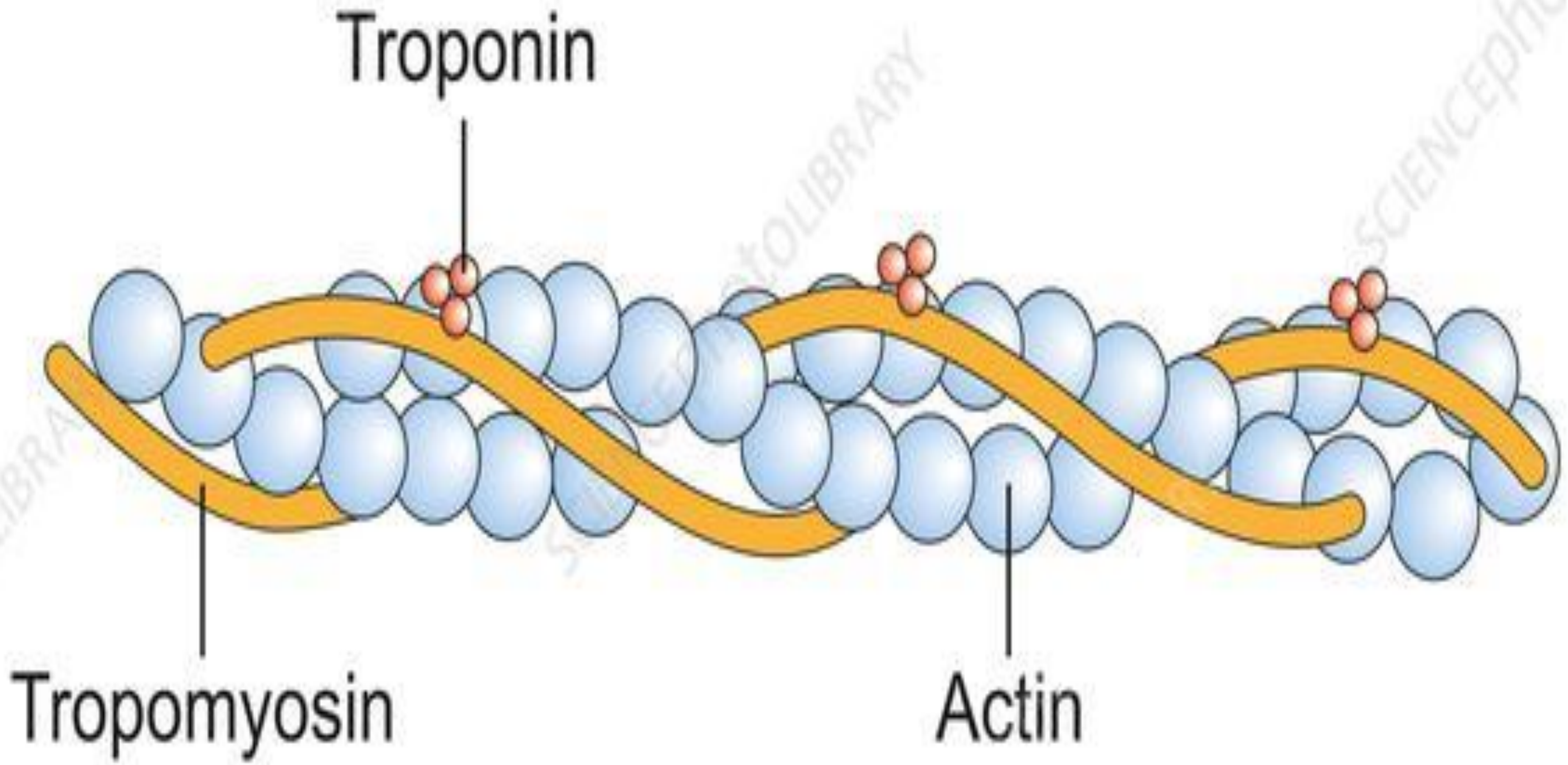
- Actin(2 G-actin molecules intermingle and form F-actin.

2 G-actin= filamentous(F) actin.

- Tropomyosin(tropomysin covers the binding sites of actin in relaxed state).
- Troponin(troponin-I, troponin-T, troponin-C)

Troponin-I has strong affinity for actin molecule, troponin-T binds to tropomyosin & troponin-C binds to calcium(Ca^{+2}).

Actin Filament Structure



Myosin(Thick filament)

- It is composed of 6 chains(2 heavy chains and 4 light chains).
- Two heavy chains intermingle and form the tail part of myosin filament.
- At one end, the two heavy heavy chains become separate and form two parts.
- Two light chains are attached to each part of heavy chains.
- This forms the head part of myosin filament.
- The head part of myosin filament binds with actin filament during contraction.
- The head part of myosin filament also has a special feature(ATPase activity).

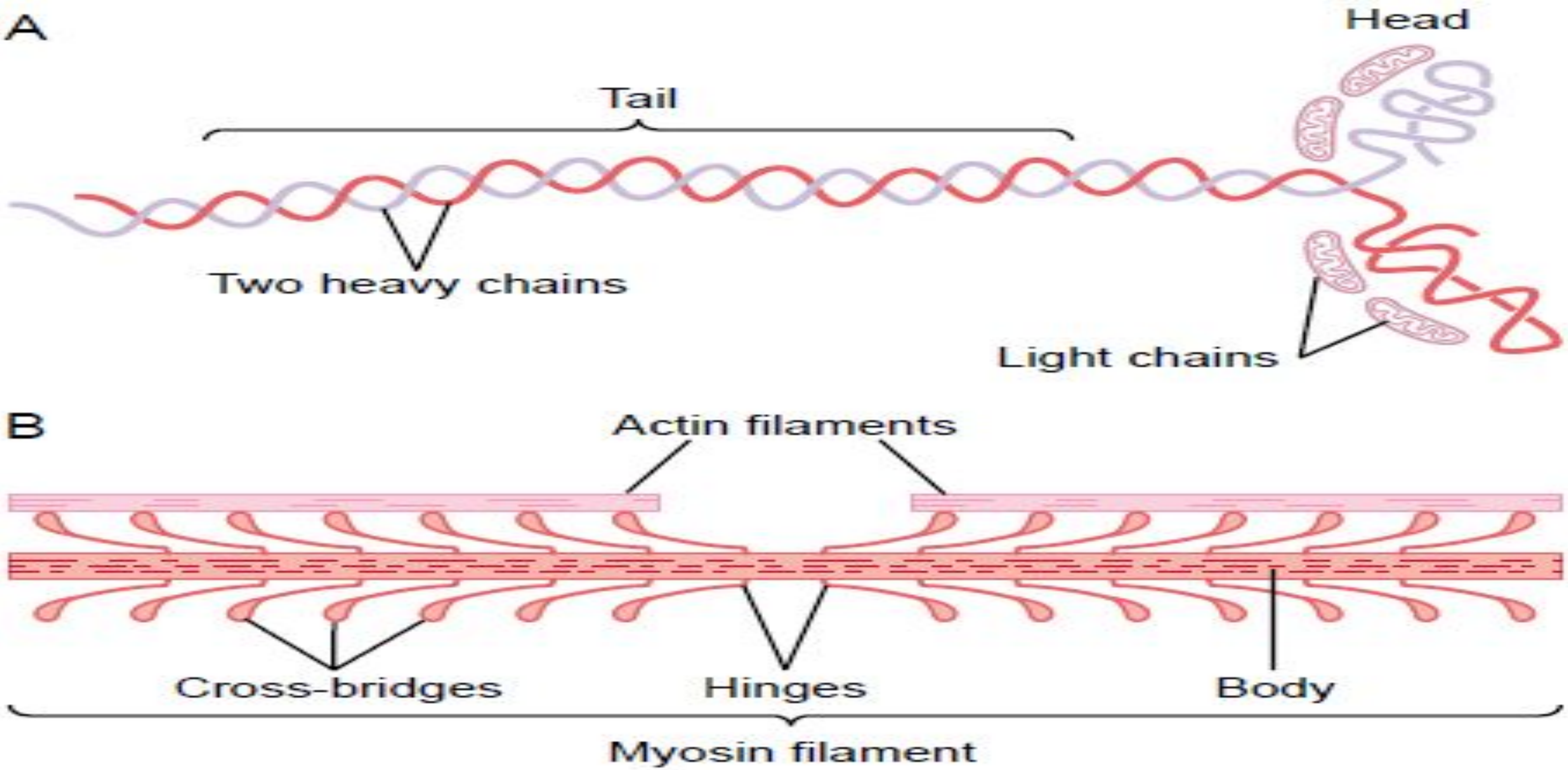


Figure 6–5

A, Myosin molecule. *B*, Combination of many myosin molecules to form a myosin filament. Also shown are thousands of myosin *cross-bridges* and interaction between the *heads* of the cross-bridges with adjacent actin filaments.

Motor unit

A motor unit is the functional unit of movement in neuromuscular system.

It consists of:

- Motor neuron
- Muscle fibre it innervates.

How motor unit works?

1. The motor neuron generates an electrical impulse.
2. Impulse travels down its axon terminal.
3. Reaches the muscle fibres at neuromuscular junction(NMJ).

Small motor unit:

- Few muscle fibres per neuron.
- Precise muscle control.
- Found in muscles like those controlling eye movements.

Large motor unit:

- Hundred to thousands of fibres per neuron.
- Produce powerful contractions
- Found in large muscles like quadriceps.

Neuromuscular transmission

Definition:

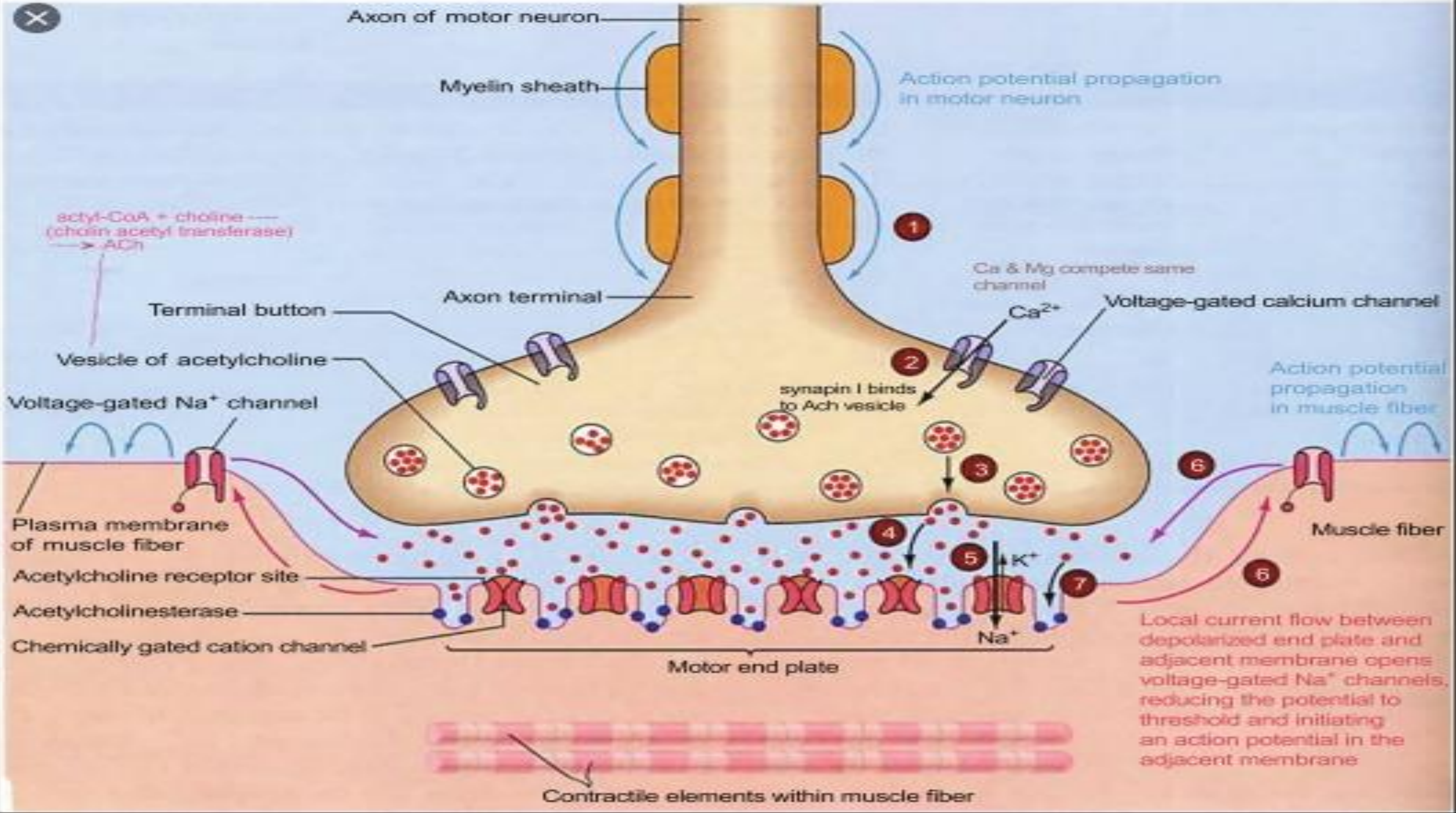
Neuromuscular transmission is a process by which a nerve impulse is transmitted from a motor neuron to skeletal muscle fibre leading to muscle excitation.

Steps of neuromuscular transmission

- Arrival of nerve impulse at neuromuscular junction
- Calcium(Ca^{+2}) influx
- Release of Acetylcholine

Cont..

- Release of Acetylcholine(ACh)
- Binding of ACh to Nm receptors
- Production of End Plate Potential (EPP)
- Muscle Action Potential (MAP)
- Termination of signal



Axon of motor neuron
Myelin sheath
Action potential propagation in motor neuron
actyl-CoA + choline ---- (choin acetyl transferase) ----> ACh
1
Ca & Mg compete same channel
Axon terminal
Terminal button
Voltage-gated calcium channel
Ca^{2+}
Vesicle of acetylcholine
2
synapin I binds to Ach vesicle
Voltage-gated Na^+ channel
Action potential propagation in muscle fiber
3
6
Plasma membrane of muscle fiber
4
Muscle fiber
5
K^+
Acetylcholine receptor site
7
6
Acetylcholinesterase
Na^+
Chemically gated cation channel
Local current flow between depolarized end plate and adjacent membrane opens voltage-gated Na^+ channels, reducing the potential to threshold and initiating an action potential in the adjacent membrane
Motor end plate
Contractile elements within muscle fiber

Excitation-Contraction Coupling (ECC)

Muscle contraction does not occur immediately after a nerve impulse.There is a critical linking process that converts:

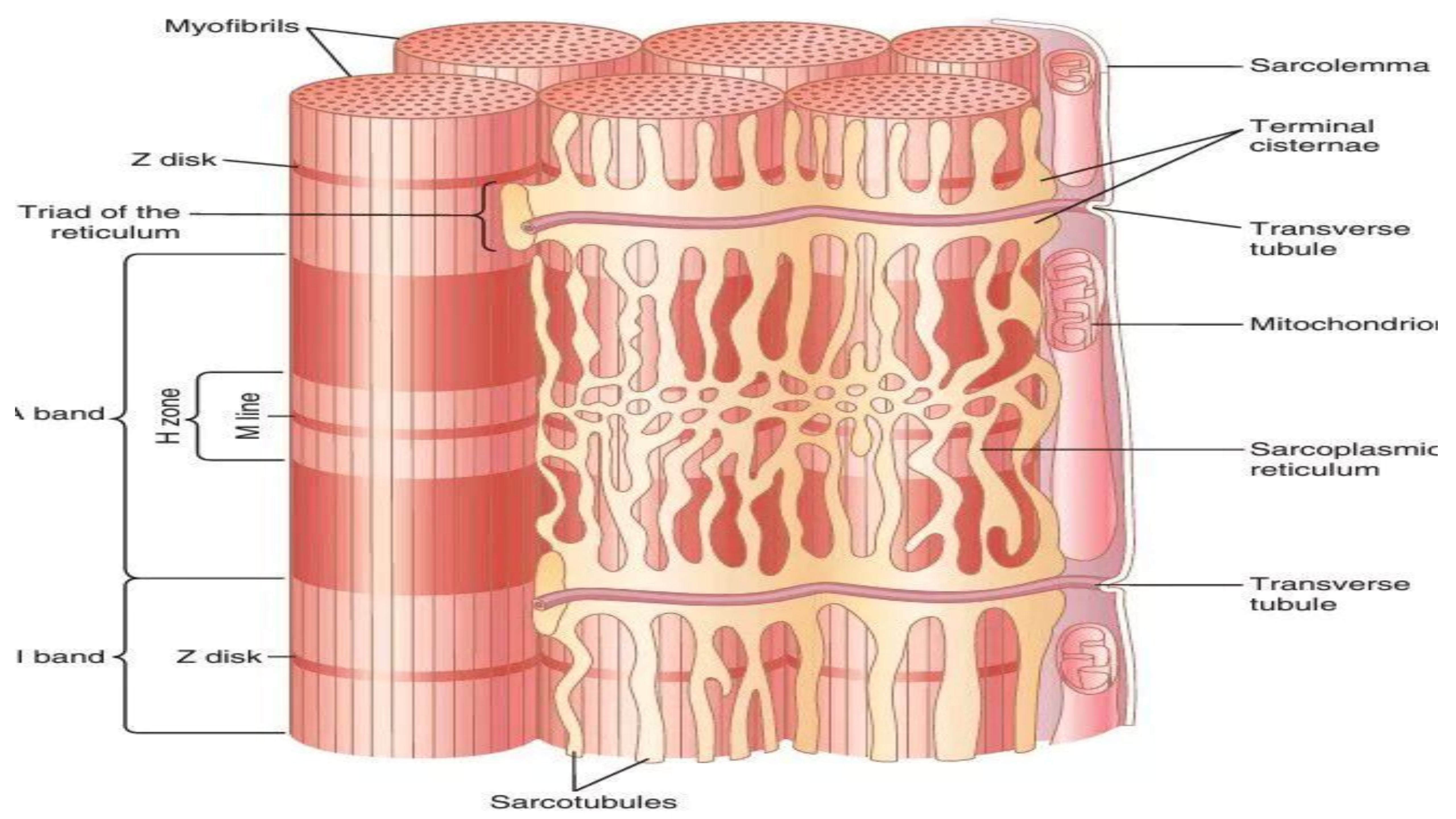
Electrical excitation - > Mechanical contraction

Definition

Excitation-contraction coupling is the sequence of events by which an action potential in the muscle fibre leads to muscle contraction.

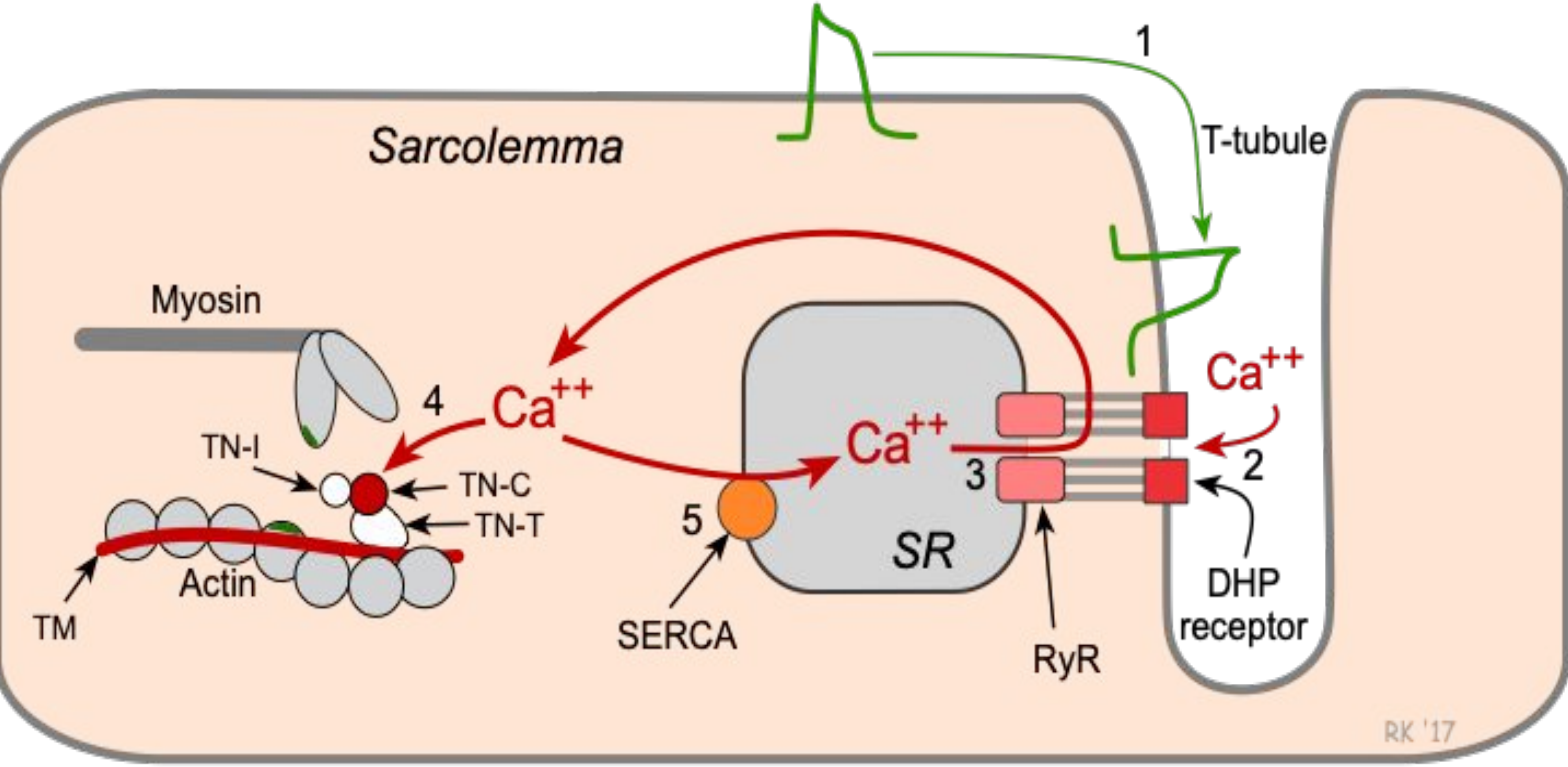
Structures involved in excitation-contraction coupling (ECC)

- Sarcolemma
- T-tubules
- Sarcoplasmic reticulum (SR)
- Myofibrils (Actin-Myosin)


Myofibrils
Sarcolemma
Terminal cisternae
Z disk
Triad of the reticulum
Transverse tubule
Mitochondrion
A band
H zone
M line
Sarcoplasmic reticulum
Transverse tubule
I band
Z disk
Sarcotubules

Mechanism of Excitation-Contraction Coupling (ECC)

- Excitation at neuromuscular junction(NMJ)
- Generation of muscle action potential(MAP)
- Activation of RyR(Ryanodine receptors)
- Calcium(Ca^{+2}) release from sarcoplasmic reticulum(SR)
- Binding of calcium(Ca^{+2}) to troponin-C
- Actin-Myosin interaction(cross-bridge formation) and muscle contraction.


1
Sarcolemma
T-tubule
Myosin
4
Ca++
Ca++
Ca++
TN-I
TN-C
TN-T
3
2
5
SR
Actin
TM
SERCA
RyR
DHP
receptor
RK '17

Step#1: Excitation at NMJ

When a motor neuron action potential reaches the axon terminal it causes the opening of voltage-gated calcium(Ca^{+2}) channels.Calcium(Ca^{+2}) enters the nerve terminal and ACh is released in the synaptic cleft which then opens the sodium(Na^{+}) channels and a local action potential is generated at NMJ.

Step#2: Generation of muscle-action potential(MAP)

After the generation of End-Plate potential(EPP),if threshold is achieved this action potential spreads over sarcolemma and T-tubules.

Important proteins

- ***Dihydropyridine receptors(DHPR)***

These are the voltage-gated receptors in the membrane of T-tubules.

- ***Ryanodine receptors(RyR)***

These are the calcium release channels in the membrane of sarcoplasmic reticulum.

Step#3: Activation of Ryanodine receptors(RyR)

As the muscle action potential(MAP) travels along the T-tubules, dihydropyridine receptors(DHPR) sense the voltage change and mechanically activates ryanodine receptors(RyR).

Step#4: Release of calcium(Ca^{+2})

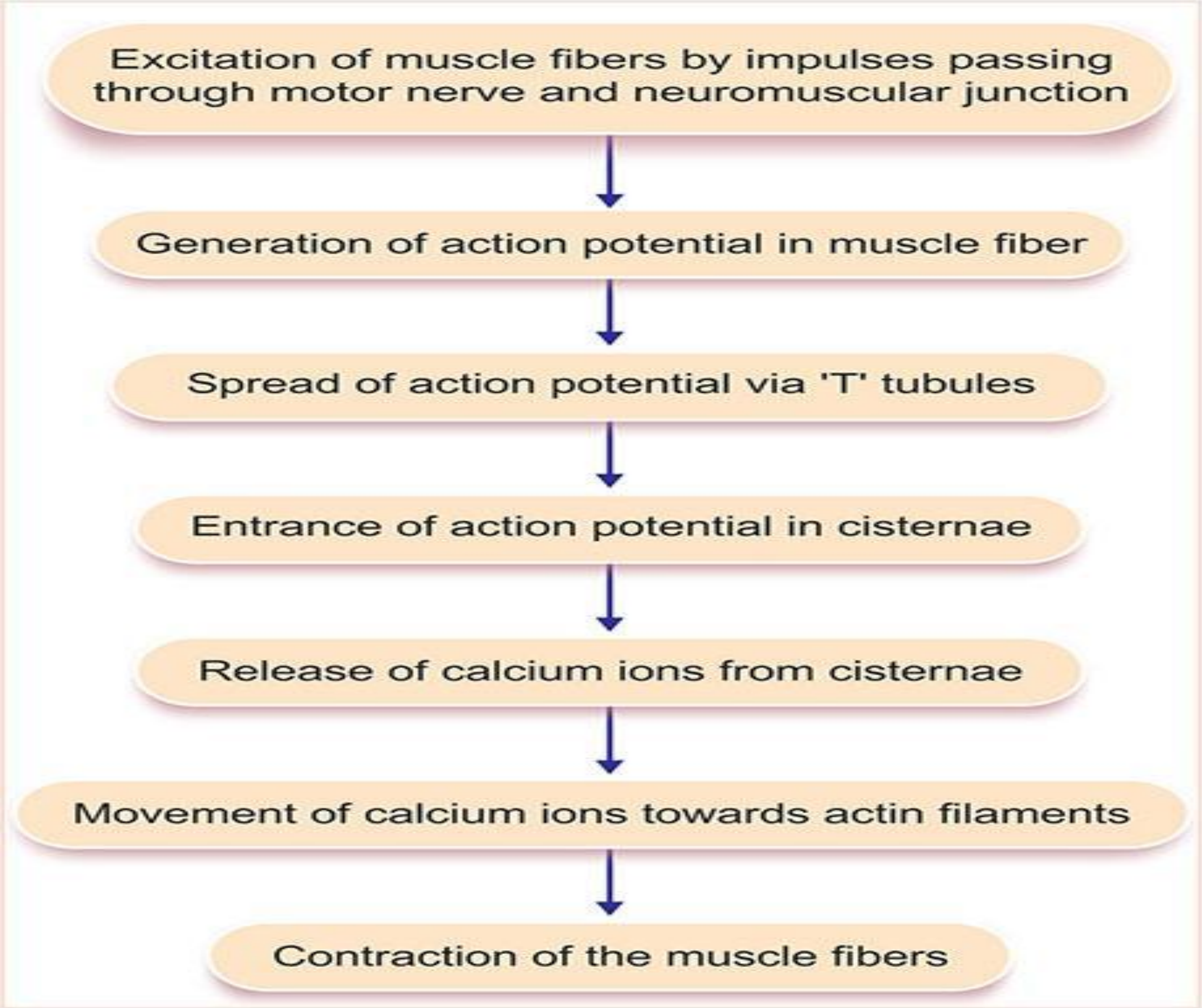
Activated ryanodine receptors(RyR) causes the release of calcium ions(Ca^{+2}) from sarcoplasmic reticulum into the sarcoplasm.

Step#5: Binding of calcium(Ca^{+2}) to troponin-C

Calcium(Ca^{+2}) released from SR into the sarcoplasm then binds to troponin-C due to which troponin changes its shape and tropomyosin moves away from the actin binding sites.

Step#6: Actin-Myosin interaction(muscle contraction

Myosin head binds to the free actin sites and cross-bridge interaction b/w actin and myosin is formed.Th cross-bridge interaction leads to the sliding of actin over myosin.This sliding of actin over myosin is called ***POWER STROKE***.


Excitation of muscle fibers by impulses passing through motor nerve and neuromuscular junction
Generation of action potential in muscle fiber
Spread of action potential via 'T' tubules
Entrance of action potential in cisternae
Release of calcium ions from cisternae
Movement of calcium ions towards actin filaments
Contraction of the muscle fibers

Clinical correlation of Excitation-Contraction Coupling(ECC)

Malignant Hyperthermia

Defective ryanodine receptors(RyR) causes excessive release of calcium(Ca^{+2}) which results in sustained contractions and hyperthermia.

Quick review

Nerve AP -> ACh release -> End Plate Potential -> Muscle Action Potential -> Calcium(Ca^{+2}) release -> Cross-bridge development -> Muscle contraction.

Muscle tone

Definition:

Muscle tone is the continuous, passive, partially contracted state of skeletal muscles even when the body is at rest.

BASELINE TENSION -> MUSCLE TONE

- Present all the time.
- Muscles are not completely relaxed.
- Small number of muscle fibres are contracting continuously.

Why muscle tone is important?

- Muscle tone helps to maintain posture(standing, sitting upright).
- It stabilize the joints.
- Keep muscles ready for voluntary movement.
- Prevent excessive stretching of muscles.

Clinical correlation

- **Hypotonia**

Mucle tone is decreased.

- **Hypertonia**

Increased muscle tone that causes spasticity/rigidity of muscles.

Post-assessment

Q1. Arrange the steps in order:

1) Ca^{+2} binds to troponin
2) ACh released
3) Action potential travels along sarcolemma
4) Myosin binds to actin
5) Ca^{+2} released from sarcoplasmic reticulum.

Q2.Tropomyosin blocks myosin-binding sites on actin when muscle is relaxed?(T/F)

Cont..

Q3. Ca^{+2} is stored in sarcolemma?(T/F)

Q4. T-tubules help transmit action potential deep into the muscle fibre?(T/F)

Q5.Muscle tone is best defined as:

a) Max. force generated during contraction.
b) Continuous partial contraction of resting muscle.
c) Muscle fatigue after repeated use of muscle.
d) Rapid alternating contractions.